International Baccalaureate®
Baccalauréat International
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BIOLOGY
HIGHER LEVEL
PAPER 3

SPECIMEN PAPER

1 hour 15 minutes

Candidate session number

Examination code

				–				

INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all questions.
- Section B: answer all of the questions from one of the options.
- Write your answers in the boxes provided.
- A calculator is required for this paper.
- The maximum mark for this examination paper is *[45 marks]*.

Option	Questions
Option A — Neurobiology and behaviour	4 – 8
Option B — Biotechnology and bioinformatics	9 – 13
Option C — Ecology and conservation	15 – 19
Option D — Human physiology	20 – 25

40EP01

40 pages

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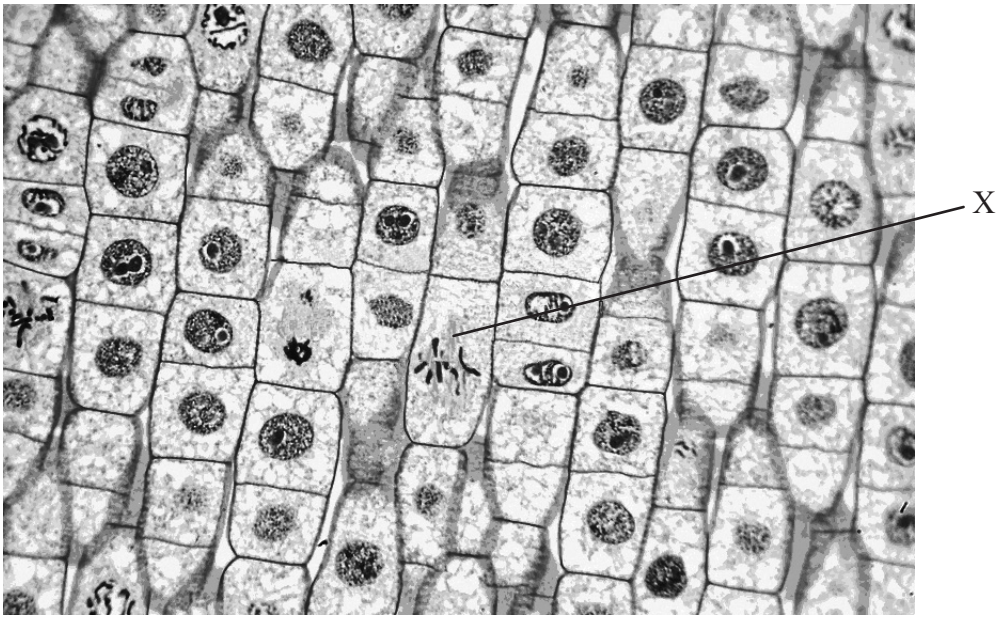
SECTION A

1. The figure is a photomicrograph of cells from an onion root tip.



[Source: adapted from http://fphoto.photoshelter.com/image/I0000AUwvOv1erSI]

(a) Identify the phase of mitosis represented by cell X. *[1]*

. . . . . .

(b) Outline **two** distinctive features of cells undergoing mitosis in the micrograph. *[2]*

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(This question continues on the following page)

(Question 1 continued)

(c) Explain how a mitotic index can be calculated from the image. *[2]*

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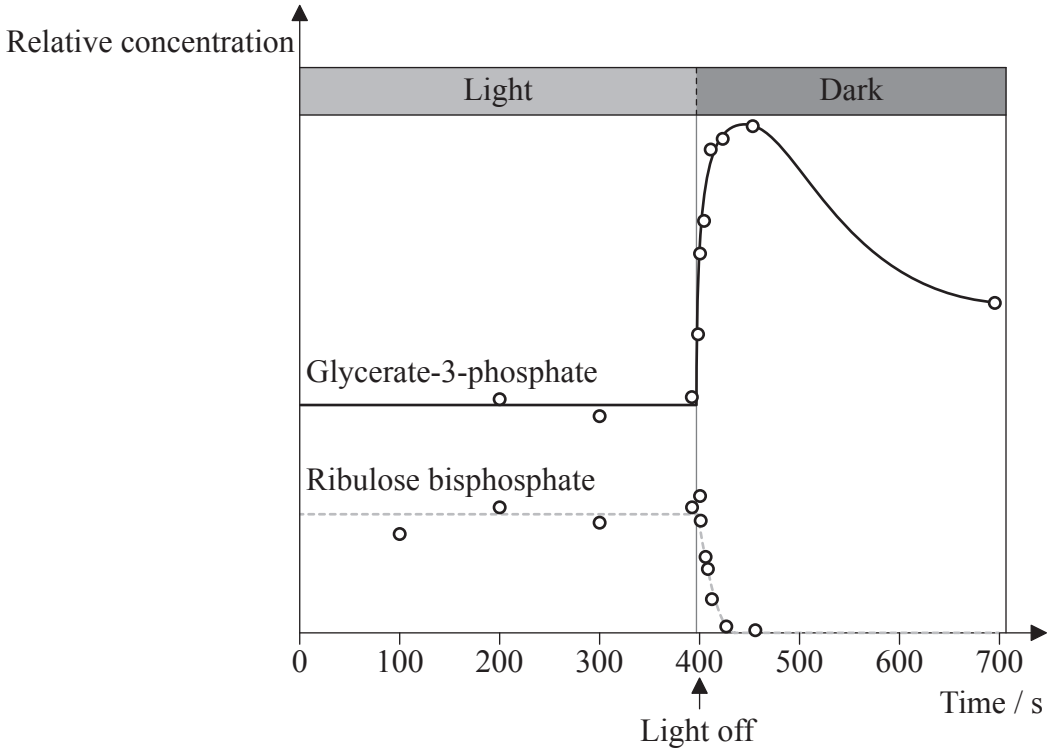
2. Concentrations of ribulose bisphosphate and glycerate-3-phosphate were monitored in a culture of cells of the alga, *Scenedesmus*. The algae were kept in bright light and then in the dark.



[Source: adapted from the course companion]

(a) Explain the change that took place in the 50 seconds after the start of darkness to the concentration of glycerate-3-phosphate. *[2]*

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(This question continues on the following page)

(Question 2 continued)

(b) Predict the effect on the concentration of ribulose bisphosphate, of turning the bright light back on after the period of darkness. *[3]*

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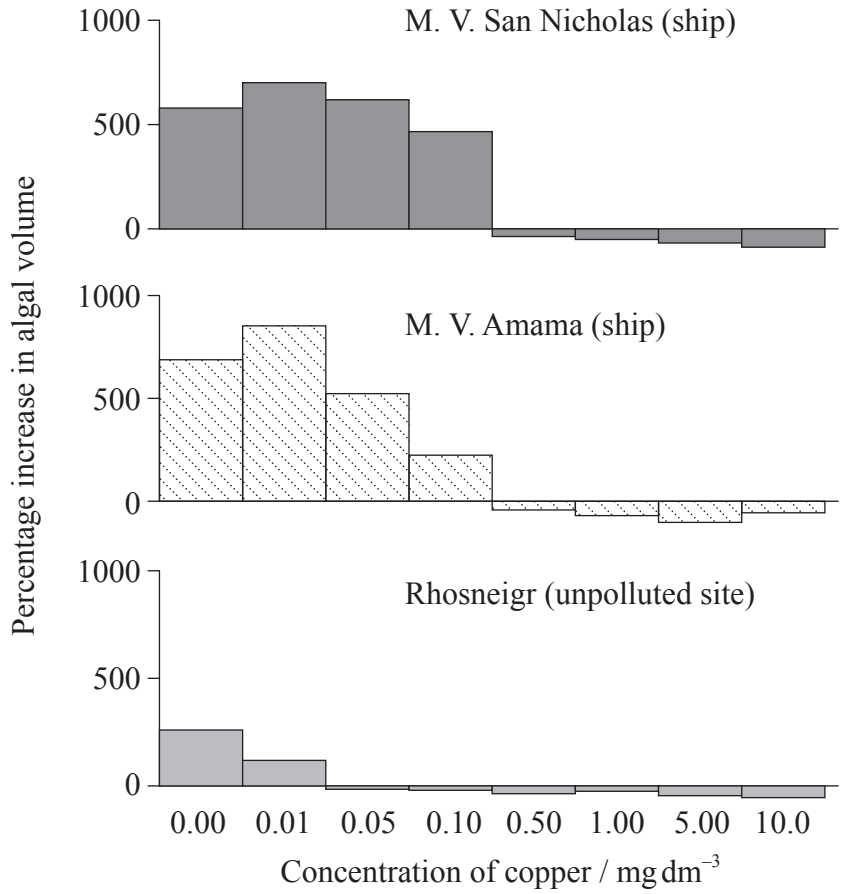
3. Samples of the alga, *Ectocarpus siliculosus*, were taken from three sites. Two samples came from the underside of ships that had been painted with a copper-containing paint. The other sample came from a nearby unpolluted environment at Rhosneigr, north Wales (UK).

The samples of algae were cultured at different copper concentrations and the changes in the number of algae were monitored. The results are shown in the bar charts. Positive values indicate an increase in numbers of algae and negative values indicate a decrease.



[Source: adapted from Russell and Morris (1970)]

(a) Outline the evidence from the graph for greater copper tolerance in the samples of algae taken from the ships than in the samples taken from the unpolluted environment. *[2]*

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(This question continues on the following page)

(Question 3 continued)

(b) Explain how copper tolerance could develop in the algae in environments polluted with copper. *[3]*

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SECTION B

Option A — Neurobiology and behaviour

4. Drug addiction is characterized by compulsive and at times uncontrollable drug craving, seeking and use.

A study was undertaken to compare ten drugs, using a scale from 0 to 3 for intensity of response, psychological addiction and physical addiction, to create a overall mean score for addiction.

Drug	Intensity of effect	Psychological addiction	Physical addiction	Overall mean
Heroin	3.0	3.0	3.0	3.00
Cocaine	3.0	2.8	1.3	2.37
Alcohol	2.3	1.9	1.6	1.93
Tobacco	2.3	2.6	1.8	2.23
Barbiturates	2.0	2.2	1.8	2.01
Benzodiazepines	1.7	2.1	1.8	1.83
Amphetamine	2.0	1.9	1.1	1.67
LSD	2.2	1.1	0.3	1.23
Cannabis	1.9	1.7	0.8	1.51
Ecstasy	1.5	1.2	0.7	1.13

[Source: adapted from D Nutt, *et al.*, (2007), *The Lancet*, **369**, pages 1047–1053]

(a) Using the table, identify which drug causes the lowest physical addiction. *[1]*

. . . . . .

(Option A continues on the following page)

(Option A, question 4 continued)

(b) (i) State **one** example of a drug that is a stimulant and **one** example that is a sedative. *[2]*

Stimulant: . . . . . .

Sedative: . . . . . .

(ii) Outline the effects of MDMA (ecstasy) on synaptic transmission. *[2]*

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(c) Novocaine is an example of a drug used as anesthetic. Describe the effects of local anesthetics on the nervous system. *[2]*

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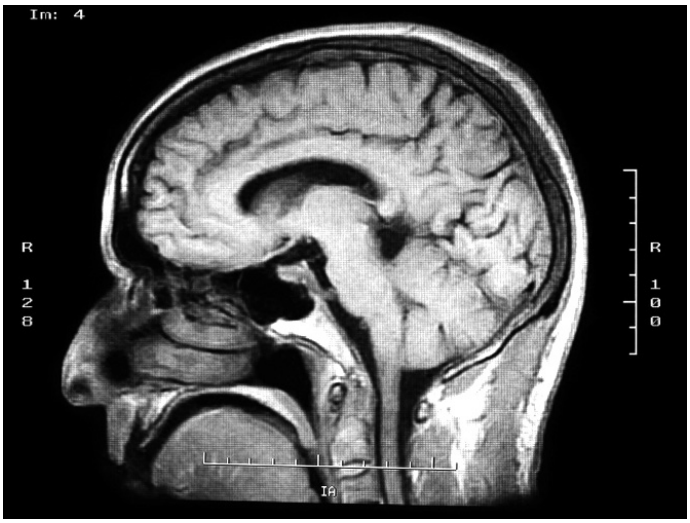
(Option A continues on the following page)

(Option A continued)

5. Brain death is a clinical diagnosis based on the absence of neurological function, with a known irreversible cause of coma.



[Source: www.npr.org/blogs/health/2012/10/16/162997951/teenage-brains-are-malleable-and-vulnerable-researchers-say, courtesy of iStock]

(a) Explain a **named** method to assess brain damage. *[2]*

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(b) Distinguish between a reflex arc and other responses by the nervous system. *[1]*

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(Option A continues on the following page)

(Option A, question 5 continued)

(c) Describe the events that occur in the nervous system when something very hot is touched. *[3]*

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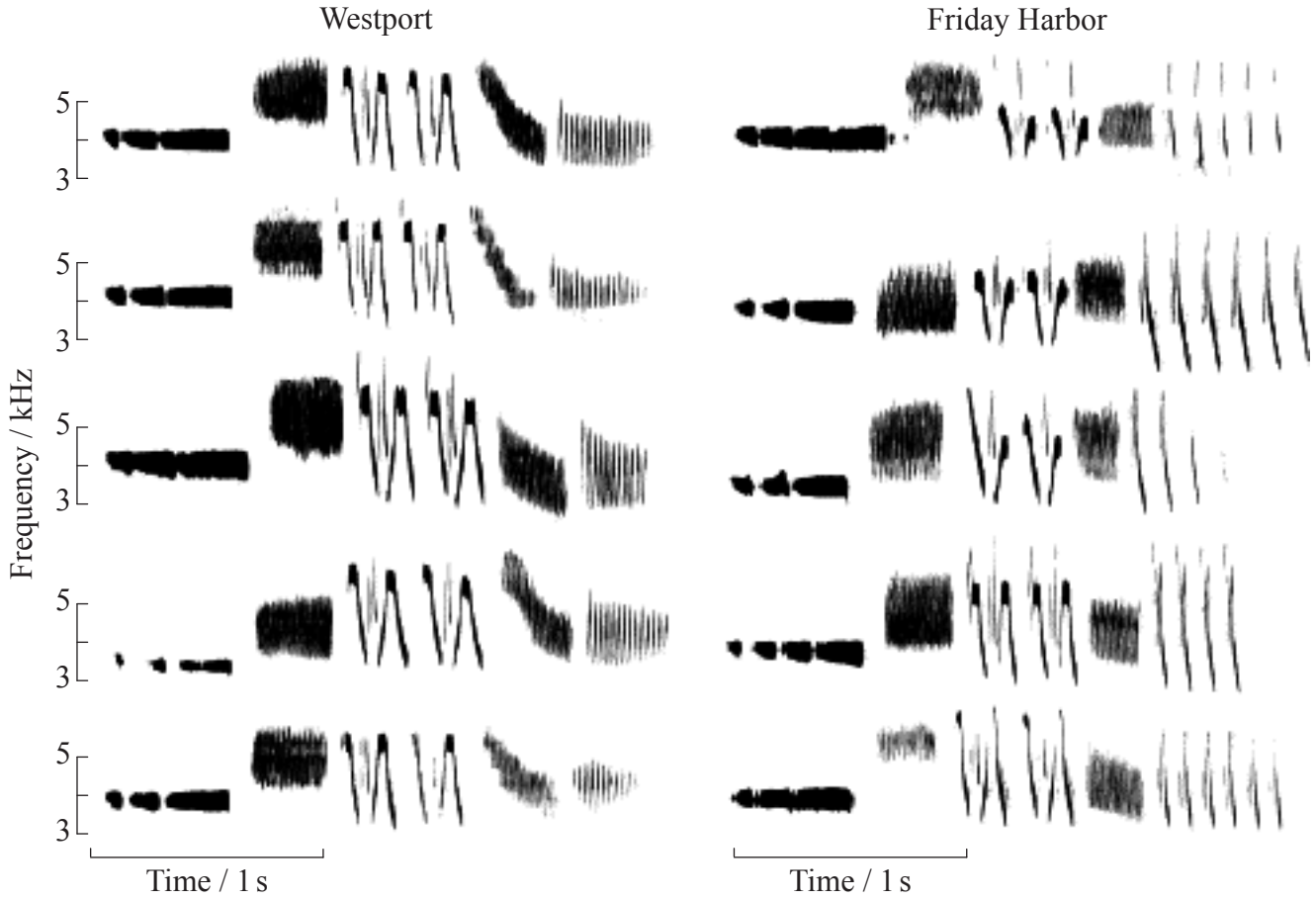
(Option A continues on the following page)

(Option A continued)

6. The sounds that birds produce can be categorized into two distinct classes: calls and songs. A song tends to be a long and complex vocalization produced during a breeding season. Songs are organized into several phrases (or motifs) which consist of a series of syllables.

The sonogram shows the birdsong of ten different male, white-crowned sparrows. Five are from Westport and five are from Friday Harbor in the Pacific Northwest coast of North America.



[Source: adapted from DA Nelson and JA Soha, (2004), *Animal Behaviour*, **68** (2), pages 395–405]

(a) Identify **one** similarity in all of the birdsongs. *[1]*

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(Option A continues on the following page)

(Option A, question 6 continued)

(b) Identify **one** difference between the birdsongs at Westport and Friday Harbor. *[1]*

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(c) Discuss the role of inheritance and learning in the development of birdsong in birds. *[3]*

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(d) Outline Pavlov's experiments into conditioning in dogs. *[3]*

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(Option A continues on the following page)

(Option A continued)

7. Explain altruistic behaviour using blood sharing in vampire bats as an example. *[3]*

[Source: http://animals.nationalgeographic.com/staticfiles/NGS/Shared/StaticFiles/animals/images/1024/vampire-bat.jpg]

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(Option A continues on the following page)

(Option A continued)

8. Explain how sound is received, transmitted and processed in humans. *[6]*

End of Option A

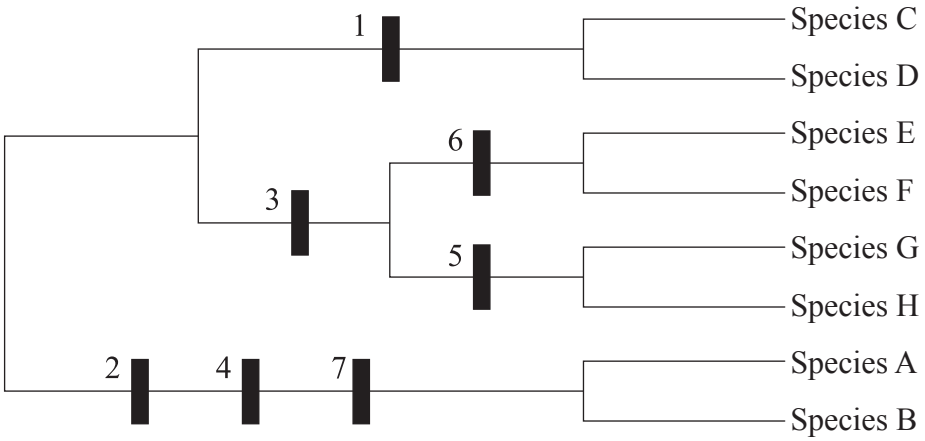
Option B — Biotechnology and bioinformatics

9. (a) State **one** difference between the p arm and the q arm of chromosome 21. *[1]*

. . . . . .

(b) The diagram shows the nucleotide alignment of an area in the DNA of eight related species and its corresponding cladogram. Numbers 1 to 7 indicate a nucleotide which has mutated.

	1	2	3	4	5	6	7
Species A	ACCAGC	**C**	TGTGCATCGATG	**A**	CGACTAAGTGATACCAT	AA	**A**GACT
Species B	ACCAGC	**C**	TGTGCATCGATG	**A**	CGACTAAGTGATACCAT	AA	**A**GACT
Species C	AC	**G**	AGCATGTGCATCGATG	CCGACTAAGTGATACCAT		AA	TGACT
Species D	AC	**G**	AGCATGTGCATCGATG	CCGACTAAGTGATACCAT		AA	TGACT
Species E	ACCAGCATGTG	**T**	ATCGATG	CCGACTAAGTGATACCA		**A**	AATGACT
Species F	ACCAGCATGTG	**T**	ATCGATG	CCGACTAAGTGATACCA		**A**	AATGACT
Species G	ACCAGCATGTG	**T**	ATCGATG	CCGACTAAGTG	**C**	TACCAT	AATGACT
Species H	ACCAGCATGTG	**T**	ATCGATG	CCGACTAAGTG	**C**	TACCAT	AATGACT



[Source: adapted from www.life.illinois.edu]

(Option B continues on the following page)

(Option B, question 9 continued)

(i) Outline the nature of mutation 6. *[2]*

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(ii) Identify a mutation which according to the cladogram, is common to species G and E. *[1]*

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(c) If scientists have determined the amino acid sequence of a novel protein, explain how they could find whether a similar protein is found in other organisms. *[2]*

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(Option B continues on the following page)

(Option B continued)

10. (a) Explain how nutrient levels affect the production of penicillin in a fermenter. *[3]*

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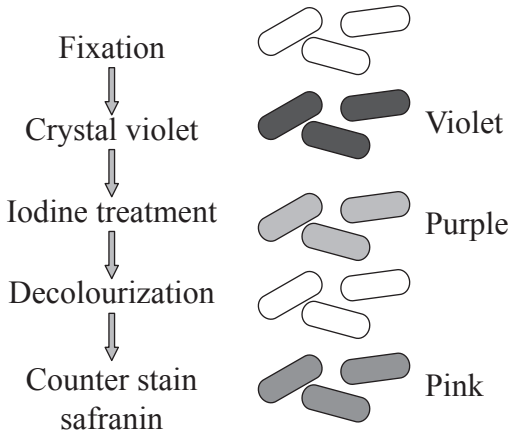
(b) The diagram represents the outcome of a Gram staining procedure.



[Source: adapted from http://pathmicro.med.sc.edu/fox/gram-st.jpg]

(i) Outline what must be done during the fixation stage. *[2]*

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(Option B continues on the following page)

(Option B, question 10 continued)

(ii) Deduce, with a reason, what type of bacterium is represented in the diagram. *[2]*

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11. (a) In order to produce a transgenic organism, other types of sequences need to be inserted into the host genome in addition to the target gene. List **two** examples of other types of sequences that have to be inserted. *[2]*

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(b) The company BASF produces a genetically modified potato called Amflora. Outline the purpose of modifying the potato. *[2]*

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(Option B continues on the following page)

(Option B continued)

12. Explain the use of biofilms in trickle filter beds for sewage treatment. *[3]*

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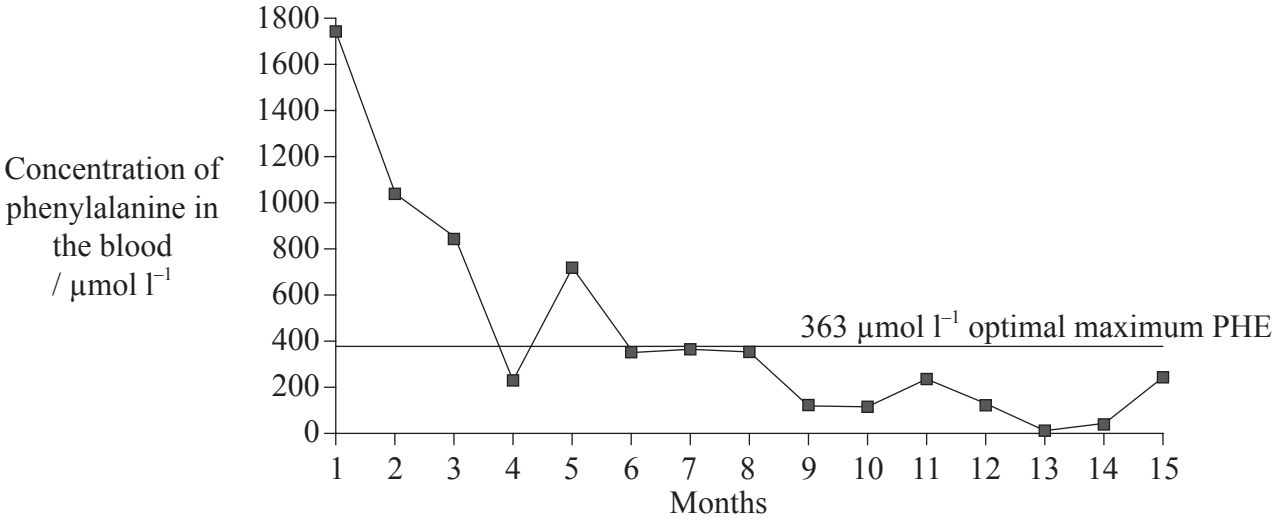
(Option B continues on the following page)

(Option B continued)

13. Phenylketonuria (PKU) is a recessive genetic disease in which the affected individual lacks functional phenylalanine hydroxylase. As a consequence, affected individuals cannot convert dietary phenylalanine (an amino acid) to the amino acid tyrosine and phenylalanine (PHE) builds up in the blood. The graph shows the levels of phenylalanine in the blood of a newborn baby with PKU until the age of 15 months. The baby was fed a modified diet to control blood phenylalanine. The target maximum blood PHE is 363 μmol l^{-1}.



[Source: www.pahdb.mcgill.ca/?Topic=Information&Section=Clinical&Page=1]

(a) Determine the number of months that the baby's blood phenylalanine level was above the optimal maximum PHE. *[1]*

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(b) Suggest how the baby's diet might have been modified. *[1]*

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(Option B continues on the following page)

(Option B continued)

(c) If a parent is a carrier for PKU, it is likely that their DNA possesses genetic markers. List **two** methods that can be used to detect the presence of these markers. *[2]*

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(Option B continues on the following page)

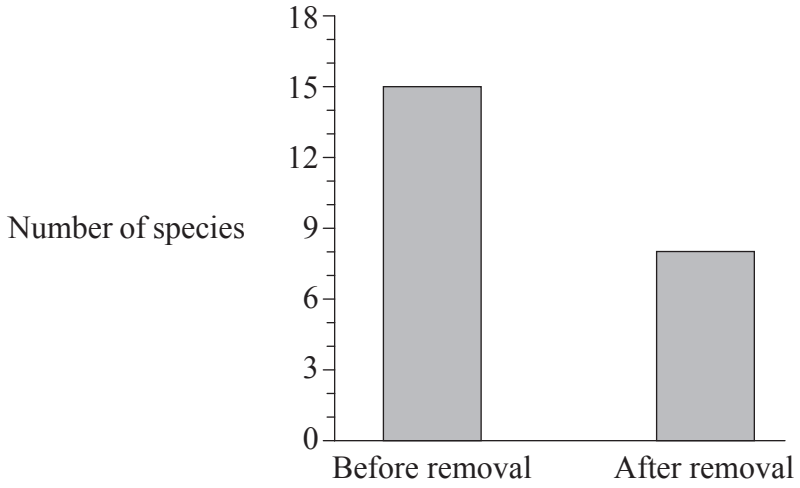
(Option B continued)

14. Outline the ELISA test for screening for HIV infection. *[6]*

End of Option B

Option C — Ecology and conservation

15. In a study examining the ecological role of keystone species, the sea star, *Pisaster*, was removed from a study area. The sea star is a top carnivore. After an extended period of time, researchers returned to analyse changes in the species diversity in the study area. The bar chart shows the species diversity before and after the removal from the habitat.



[Source: adapted from R T Paine, (1966), *The American Naturalist*, **100** (910), pages 65–75]

(a) State the effect of removing the sea star. *[1]*

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(b) (i) Define keystone species. *[1]*

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(Option C continues on the following page)

(Option C, question 15 continued)

(ii) Suggest reasons for the change in species diversity after removal of the sea star. *[3]*

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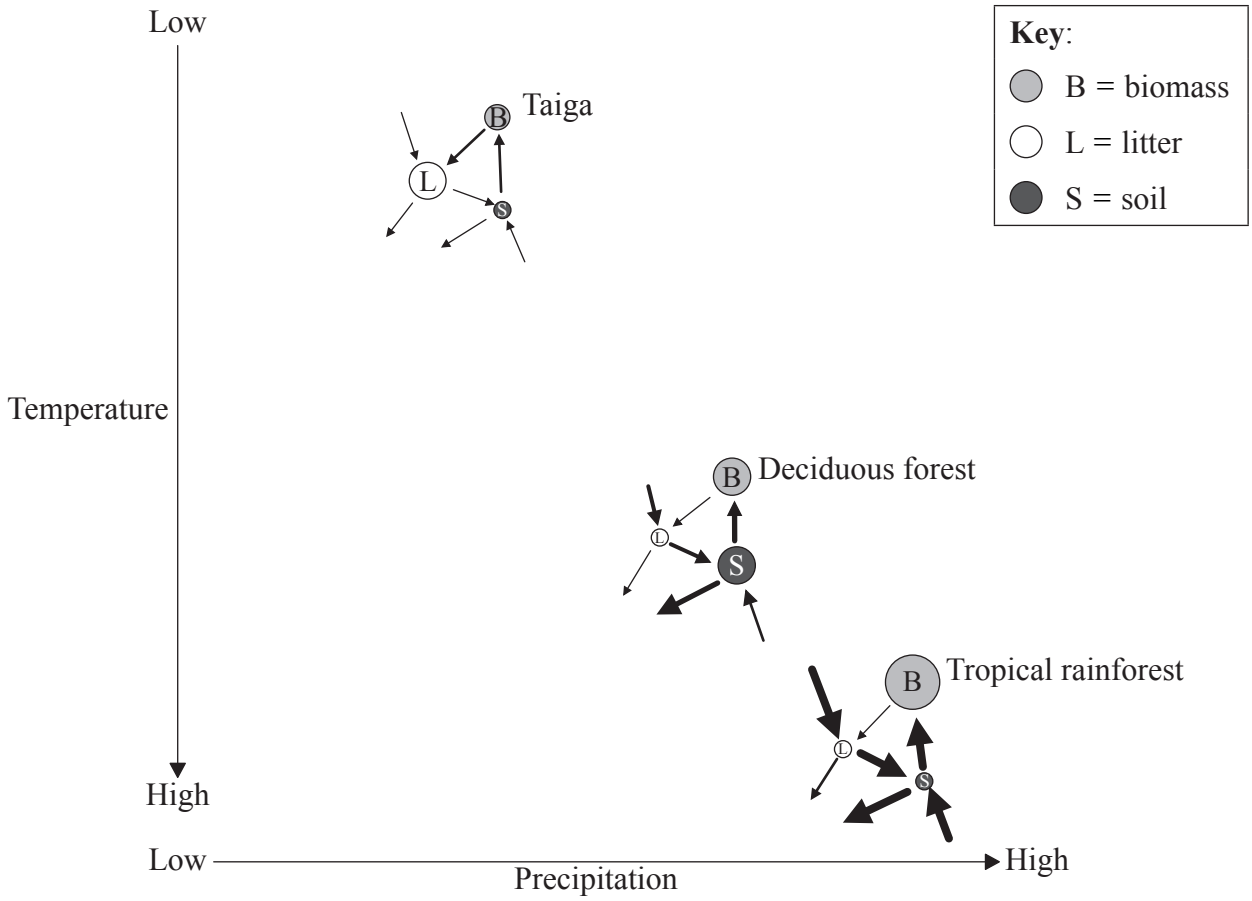
(Option C continues on the following page)

(Option C continued)

16. The model depicted represents the effect of temperature and precipitation on the flow of nutrients within different types of ecosystems. The thickness of the arrow represents the rate of nutrient flow.



[Source: adapted from www.slideshare.net/ecumene/ecosystems-3-nutrient-cycle-presentation]

(a) Identify **one** ecosystem where the litter represents the highest level of nutrient store. *[1]*

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(Option C continues on the following page)

(Option C, question 16 continued)

(b) Deduce what process is indicated by the arrow connecting the litter to the soil. *[1]*

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(c) State the relationship between the level of precipitation and the relative amount of nutrients stored in biomass. *[1]*

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(d) High levels of rainfall can lead to water-logging of soils. Outline the impact of water-logging on the nitrogen cycle. *[2]*

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(e) Deduce, with a reason, whether the models assume the ecosystems to be open **or** closed. *[2]*

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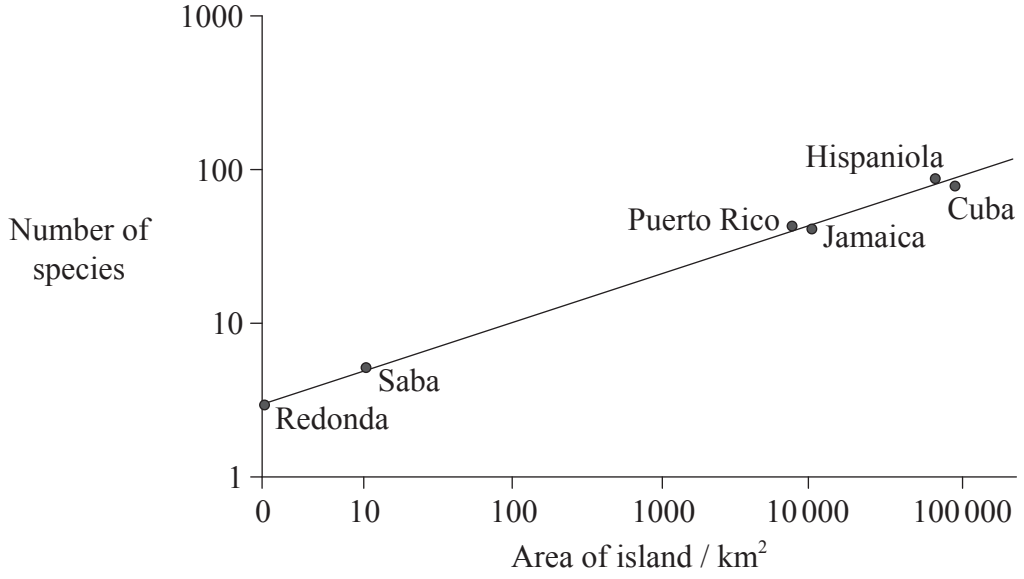
(Option C continues on the following page)

(Option C continued)

17. The graph shows the relationship between island area and reptilian diversity in the West Indies.



[Source: adapted from http://web2.uwindsor.ca/courses/biology/macisaac/55-437/lecture9.htm]

(a) Outline the relationship between island area and the number of reptile species. *[1]*

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(b) Montserrat has an area of 100 km^2. Predict the number of species of reptile that can be found there. *[1]*

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(Option C continues on the following page)

(Option C, question 17 continued)

(c) Explain how an alien species can become invasive. *[2]*

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(d) List the **two** types of data needed to calculate the value of a biotic index for a local stream. *[2]*

1. . . . . .

2. . . . . .

(Option C continues on the following page)

(Option C continued)

18. (a) With reference to **one** example of each, distinguish between a top down limiting factor and a bottom up limiting factor. *[3]*

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(b) Discuss the trade-off between the control of the malarial parasite and DDT pollution. *[3]*

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(Option C continues on the following page)

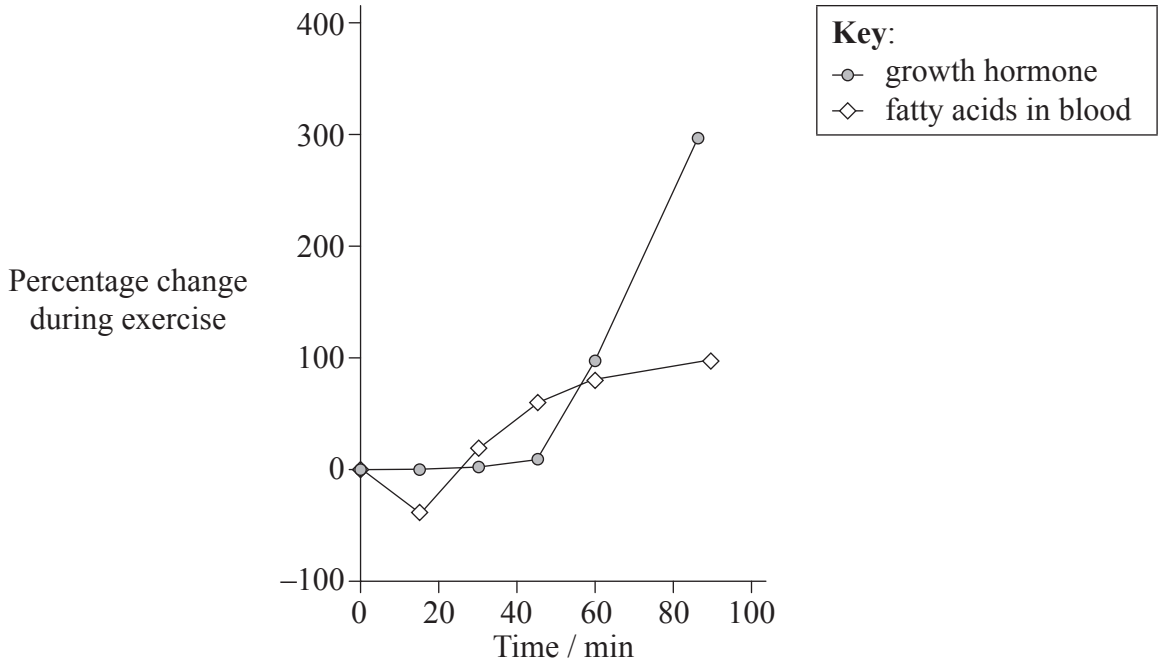
(Option C continued)

19. Evaluate the methods used to estimate the size of commercial fish stocks. *[6]*

End of Option C

Option D — Human physiology

20. The graph shows the responses in levels of growth hormone and free fatty acids to relatively intense exercise combining aerobic and anaerobic components.



[Source: adapted, by permission, from J.H. Wilmore and D.L. Costill, (2004), *Physiology of Sport and Exercise*, 3rd ed., (Champaign, IL: Human Kinetics), page 178]

(a) Identify the level of growth hormone after one hour of exercise. *[1]*

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(Option D continues on the following page)

(Option D, question 20 continued)

(b) (i) State **one** reason that motivates some athletes to take growth hormones. *[1]*

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(ii) State **one** risk associated with the use of growth hormones in sports. *[1]*

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(c) Suggest **one** possible use of fatty acids during exercise. *[1]*

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(d) Testosterone is a steroid hormone. Outline the mechanism by which steroid hormones affect target cells. *[2]*

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(Option D continues on the following page)

(Option D continued)

21. (a) The liver produces cholesterol. State **two** other functions of the liver. *[2]*

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2. . . . . .
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(b) Compare and contrast cholesterol produced by the liver and dietary cholesterol. *[2]*

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(Option D continues on the following page)

(Option D, question 21 continued)

(c) The CAT scan shows a patient who has a blocked bile duct.

[Source: adapted from http://upload.wikimedia.org/wikipedia/commons/4/4c/Obstructivebiliarydilation.png]

The blockage of the bile duct causes a build-up of bilirubin in the blood.

(i) State **one** consequence of a build-up of bilirubin in the blood. *[1]*

. . . . . .

(ii) State **one** other possible cause for the build-up of bilirubin in the blood. *[1]*

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(Option D continues on the following page)

(Option D continued)

22. The X-ray shows the legs of a young boy who suffers from rickets.

[Source: www.millathomeopathy.com/images/disease-cd-rickets.jpg]

(a) State the symptom of rickets evident in the X-ray. *[1]*

. . . . . .

(b) State the main cause of rickets. *[1]*

. . . . . .

(c) Identify **one** vitamin and **one** hormone that are based on a steroid ring. *[2]*

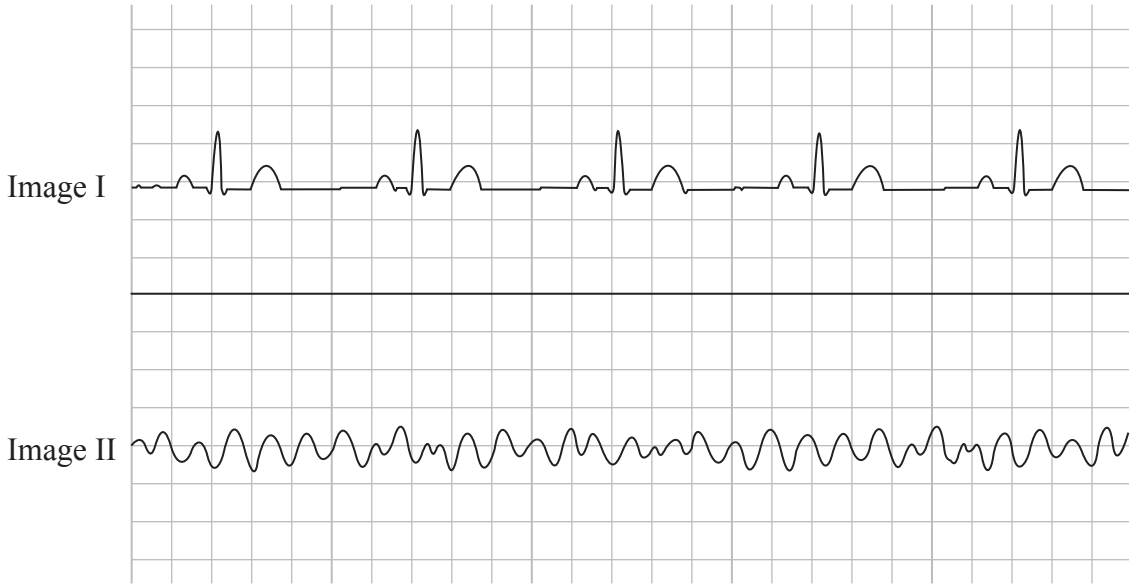
Vitamin: . . . . . .

Hormone: . . . . . .

(Option D continues on the following page)

(Option D continued)

23. Image I represents a normal heart rhythm and image II represents an abnormal heart rhythm.



[Source: adapted from www.homeheart.co.uk/ecg_example.jpg]

(a) State the name given to the abnormal rhythm pattern. *[1]*

. . . . . .

(b) State a **named** technique used to restore the normal heart rhythm. *[1]*

. . . . . .

(c) Annotate image I to indicate **one** phase where the atrium is contracting and **one** phase where the ventricle is contracting. *[2]*

(Option D continues on the following page)

(Option D, question 23 continued)

(d) State **one** unique characteristic of cardiac muscle cells. *[1]*

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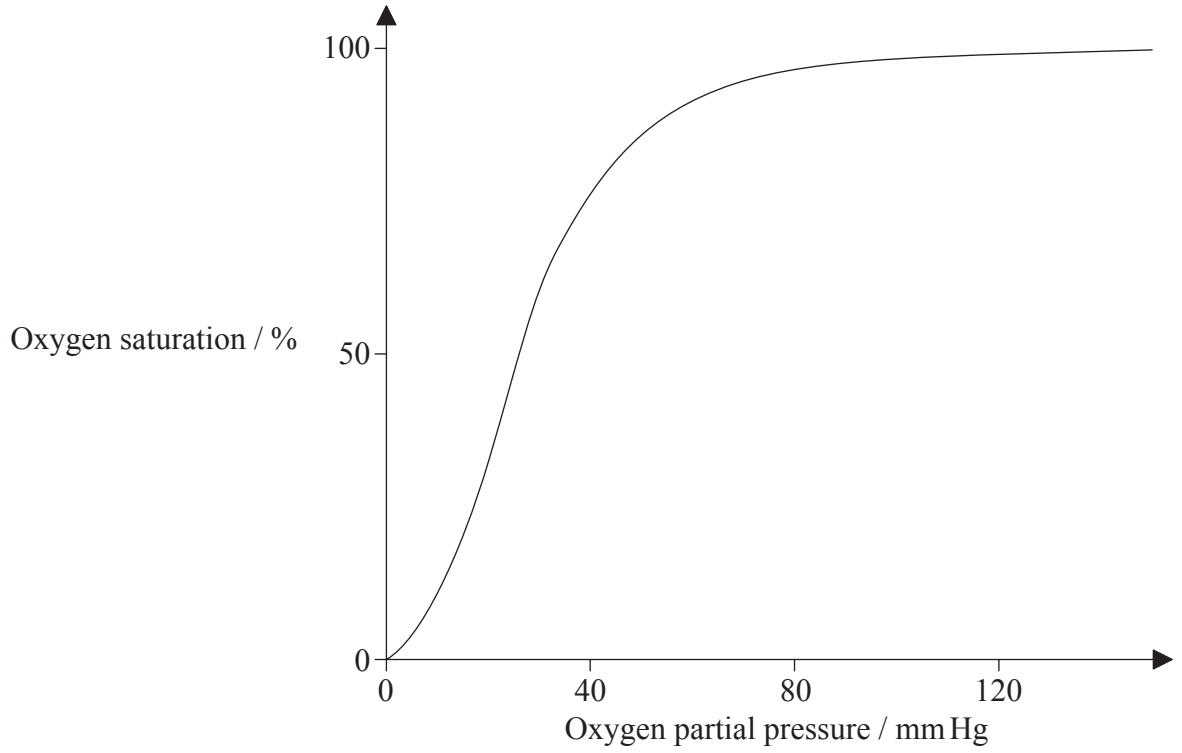
(Option D continues on the following page)

(Option D continued)

24. Gas exchange between maternal blood and fetal blood occurs in the placenta. The graph shows the dissociation curve of oxygen from the mother.



(a) State which mineral ion is found in hemoglobin. *[1]*

. . . . . .

(b) On the graph, draw the dissociation curve for fetal hemoglobin. *[2]*

(Option D continues on the following page)

(Option D continued)

25. Explain the mechanisms carried out in the blood to control the changes of pH during the transport of gases. *[6]*

End of Option D